Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State the no-air-resistance assumption.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 4 kg cart starts from rest under 24 N net force right for 3 s. Find acceleration and final velocity. State one assumption.

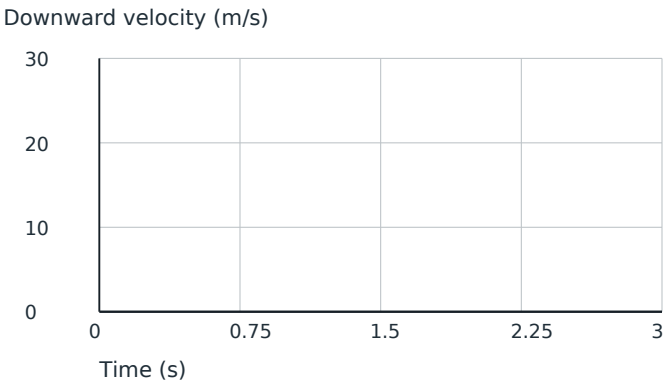
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_f = v_i + g\,\Delta t$.
Downward positive; g = 10 m/s²; gravity only.

Worked reasoning: A ball dropped from rest has downward speeds 0, 10, 20 m/s at 0, 1, 2 s in ideal free fall with g=10 m/s². Equal times give equal velocity changes.

Your turn: Sketch its velocity-time graph with downward positive. Explain why the positions are not equally spaced.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Two balls of different mass are released together in a vacuum from the same height. Compare acceleration and arrival time.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball is released from rest in a drop simulation. Ignore air resistance and use g = 10 m/s². Complete the table, then describe how the velocity changes each second.

Time (s)	Downward velocity (m/s)
0	0
1	____
2	____
3	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the claim.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A fixed-mass cart has acceleration 4 m/s² under 8 N net force. Find acceleration when net force becomes 12 N. Explain.
Use the table to keep the controlled and changed quantities distinct.

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

07 Exit check

A dropped ball falls from rest for 4 s in an air-free simulation. Use g = 10 m/s². Find its final velocity and add a correctly directed arrow to the model.

Dropped ball

State your positive direction. →

Velocity:

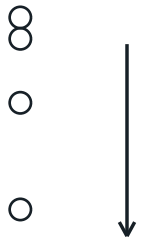
Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Draw three equal-time snapshots of a dropped ball. Then draw a heavier ball released with the same conditions in a vacuum. Compare the patterns.

Equal time intervals



Marzano check - circle: 1 2 3